PUBLIC WHITE PAPER | WP-012

Integrated Readiness Roadmap and Evidence-Gate Framework

A practical control map from technology claim to repeatable, transferable business capability

PURPOSE

Connect the Rapid Organic public white-paper series into one evidence-led programme for validation, pilot delivery, productization, deployment and responsible scale-up.

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Executive summary

Readiness is not the number of reports produced. It is the degree to which important claims are supported by controlled, reproducible and decision-relevant evidence. This framework joins the preceding white papers into one operating system with explicit gates, owners and stop conditions.

Principle	Required practice	Decision value
Claims before activities	State the claim and acceptance criterion before testing or piloting.	Prevents activity from being mistaken for proof.
Configuration control	Tie evidence to feedstock, machine, software, method and site.	Defines where a result is valid.
Independent evidence	Use qualified third parties or witnesses where claim risk warrants it.	Raises buyer and customer confidence.
Cross-functional gates	Technical, regulatory, commercial, safety and delivery evidence mature together.	Prevents one strong dimension hiding another weak one.
Honest limitations	Record uncertainty, exclusions, failures and open actions.	Makes diligence faster and more credible.

CORE CONCLUSION

The objective is not to appear ready. It is to know what is ready, what is not, what evidence closes each gap and who has authority to proceed.

1. How to use the white-paper series

The series is a connected control architecture. Each paper addresses a different failure mode; none should be treated as proof that the described controls have already been completed.

Paper	Primary question	Programme use
WP-001	What is the technology and value proposition?	Claim map and system boundary.
WP-002	How does the seafood use case work?	Sector application hypothesis.
WP-003	Can a customer build a defensible business case?	Economic model and assumptions.
WP-004	How will performance claims be validated?	Technical evidence architecture.
WP-005	What is the output and its lawful pathway?	Characterization and regulatory route.
WP-006	Can a flagship pilot prove customer value?	Commercial-validation design.
WP-007	Can units be built repeatedly and economically?	Productization and manufacturing.
WP-008	Are knowledge, data and ownership defensible?	IP and data governance.
WP-009	Can an installed fleet be supported reliably?	Service and lifecycle evidence.
WP-010	Can sites be installed and commissioned consistently?	Deployment readiness.
WP-011	Are safety and conformity risks controlled?	Safety evidence and release controls.

2. Evidence hierarchy

Evidence strength should match the consequence of the decision. A brochure, observation or successful demonstration may support exploration, but not necessarily investment, certification, scale-up or a customer guarantee.

Level	Evidence form	Permitted interpretation
E0 — assertion	Unverified description, estimate or supplier statement.	Hypothesis only.
E1 — observed	Documented run or field observation without full controls.	Feasibility signal.
E2 — controlled	Approved method, calibrated instruments, traceable configuration and raw data.	Defined performance claim.
E3 — independent	Qualified external laboratory, witness or specialist confirms result.	Higher-confidence external claim.
E4 — repeated	Result reproduced across time, batches, operators or representative sites.	Repeatability and operating-window claim.
E5 — scaled	Multiple controlled deployments show delivery, reliability and economics.	Scale and transferability claim.

RULE

A higher evidence level does not automatically broaden scope. Repetition on one feedstock does not prove performance on every organic-waste stream.

3. Integrated readiness dimensions

A programme advances only when all dimensions required for the next decision reach the defined threshold. Overall readiness should not be calculated as a misleading average.

Dimension	Key question	Typical evidence
Technical	Does the controlled configuration meet stated performance criteria?	Protocols, raw data, mass/energy balance and reports.
Output / regulatory	Is the output characterized and is the intended use lawful?	Laboratory results, classification and regulator pathway.
Commercial	Does a real customer receive measurable value and accept the model?	Baseline, pilot results, economics and conversion evidence.
Product / manufacturing	Can equivalent units be built, tested and costed?	BOM, drawings, supplier controls, FAT and cost roll-up.
Safety / conformity	Are critical risks reduced and required approvals planned or obtained?	Risk file, tests, inspections and approval records.
Deployment / service	Can a site be commissioned and supported without founder dependence?	SAT, training, spares, uptime and service records.
IP / data	Are ownership, access and defensibility controlled?	Agreements, registers, software baseline and data rights.

4. Stage model and gate logic

The stage model converts an open-ended development effort into bounded decisions. Gate approval should mean that the next expenditure and exposure are justified—not that every possible uncertainty has disappeared.

Stage	Gate decision	Minimum outcome
R0 — define	Is the claim, use case and system boundary worth testing?	Approved claim register and gap map.
R1 — prove	Has the technology met controlled technical criteria?	Configuration-specific validation evidence.
R2 — qualify output	Is composition known and intended disposition defined?	Output dossier and lawful pathway plan.
R3 — validate customer	Has a representative pilot demonstrated measurable customer value?	Signed baseline, results and closeout decision.
R4 — productize	Can a safe, controlled unit be reproduced and released?	Product baseline, suppliers, QMS and conformity route.
R5 — deploy	Can installation, commissioning and service be repeated?	SAT, handover and early-life reliability evidence.
R6 — scale / transfer	Does the system support a larger fleet, partner or transaction?	Repeated economics, fleet data and diligence pack.

5. Gate dossier and decision record

Every gate should use the same disciplined structure. This makes the programme auditable and prevents important caveats from disappearing between technical, commercial and management teams.

Dossier element	Required content
Decision requested	Proceed, proceed with conditions, repeat, redirect or stop.
Scope	Configuration, feedstock, site, operating window and time period.
Claims / criteria	Predefined measurable statements and acceptance thresholds.
Evidence index	Controlled records, raw data, reports and external confirmations.
Exceptions	Failures, deviations, missing data, limitations and temporary controls.
Risk update	New, changed and retired risks with owners and due dates.
Resources	Budget, people, suppliers, equipment and schedule for the next stage.
Approval	Named decision authority, date, conditions and expiry/review trigger.

6. Claim-to-evidence traceability

A traceability matrix connects external language to the exact evidence that supports it. This is especially important where the same result appears in sales material, grant applications, technical reports and buyer diligence.

Field	Example control question
Claim ID / wording	What exactly is being said, without promotional ambiguity?
Decision / audience	Who will rely on the claim and for what decision?
Scope / exclusions	Which materials, machine revision, conditions and uses are covered?
Metric / criterion	How is success measured and what threshold applies?
Method / evidence	Which approved protocol, raw dataset and report support it?
Evidence strength	Observed, controlled, independent, repeated or scaled?
Owner / approver	Who maintains and authorizes the claim?
Review trigger	What change, failure or time interval requires reassessment?

PUBLICATION CONTROL

If the evidence is narrower than the proposed wording, narrow the wording. Do not broaden the interpretation.

7. Configuration and data integrity

Evidence is useful only when the tested system can be reconstructed. A result without configuration and provenance may be informative, but it is weak acquisition-grade proof.

Control	Minimum record
Machine identity	Serial number, hardware revision, drawings and deviations.
Software / settings	Released version, parameter set, permissions and change log.
Feedstock	Source, sampling, composition, contaminants and preparation.
Operating conditions	Load, duration, temperature, utilities, environment and interventions.
Measurement	Instrument identity, calibration, method, timestamps and operator.
Raw data	Immutable source files, units, missing values and transformation history.
Output samples	Sample ID, custody, storage, laboratory method and result.
Approval state	Draft, reviewed, approved, superseded or invalidated.

8. Integrated pilot gate

The flagship pilot is the programme’s convergence point. It should test technical performance, customer value, output handling, operability, safety, service and data capture under one pre-agreed protocol.

Before start	Exit evidence
Representative site and feedstock selected	Scope and representativeness documented.
Baseline and counterfactual agreed	Avoided cost, labour, waste volume and current route signed.
Machine and site released	FAT, site readiness, safety actions and approvals complete.
Output route authorized	Sampling, storage, transport and intended disposition controlled.
KPI and data rights contracted	Raw data access, publication rights and confidentiality clear.
Stop / escalation rules defined	Safety, performance, odour, contamination and downtime limits.
Commercial closeout planned	Decision date, pricing path and case-study permissions set.

9. Productization and deployment gate

A successful pilot can still depend on a unique prototype and founder intervention. Product release requires evidence that build, installation, operation and service can be transferred to competent people using controlled information.

Release area	Evidence threshold
Product baseline	Approved EBOM/MBOM/SBOM, drawings, software and serial control.
Supply chain	Qualified critical suppliers, inspection criteria and alternate strategy.
Quality	Build travellers, nonconformance control, calibrated tools and release authority.
Safety / conformity	Risk controls verified and applicable approval pathway satisfied.
Factory acceptance	Every critical function passes configuration-linked FAT.
Site deployment	Site survey, utilities, logistics, SAT and handover are repeatable.
Service readiness	Spares, diagnostics, escalation, training and response model exist.
Economics / capacity	Cost roll-up, labour content, lead time and constrained capacity known.

10. Reliability and fleet-learning gate

Scale should follow demonstrated control of early-life failures. Fleet data must distinguish scheduled unavailability, customer-caused interruption and product failure using stable definitions.

Metric / record	Decision use
Availability / uptime	Confirms usable service during the agreed required period.
Failure taxonomy	Separates design, component, process, site and use causes.
MTTR / response	Shows serviceability, parts access and support performance.
Recurring defects	Triggers containment, root cause and fleet-wide CAPA.
Consumables / wear	Improves maintenance intervals and lifecycle economics.
Operator interventions	Reveals hidden labour and usability problems.
Warranty / service cost	Tests whether the commercial model remains viable.
Configuration comparison	Prevents mixed fleet data from masking revision effects.

11. Economics and commercial proof

A credible model reconciles customer savings, Rapid Organic revenue, delivery cost, risk and working capital. Technical success is necessary, but it cannot substitute for a customer decision at an acceptable price.

Economic layer	Required evidence
Customer baseline	Current hauling, tipping, handling, storage, compliance and labour costs.
Operating cost	Energy, labour, consumables, maintenance, testing and output handling.
Capital / commercial model	Purchase, lease, service, financing and installation assumptions.
Benefit realization	Measured avoided cost or value—not only modeled potential.
Sensitivity	Feedstock volume, utilization, downtime, disposal fee and energy price.
Rapid Organic margin	Delivered COGS, warranty, support, channel and overhead implications.
Conversion evidence	Paid pilot, order, contract, renewal or documented no-go reason.

COMMERCIAL BOUNDARY

A free pilot with positive feedback is useful learning, but it is not the same as willingness to pay or repeatable product-market fit.

12. Risk and issue governance

Risk registers should drive decisions, not become archives. Each material issue needs a trigger, owner, current exposure, action, deadline and evidence-based closure criterion.

Risk class	Typical gate blocker
Safety	Unverified critical safeguard or unsafe site interface.
Regulatory	No defined lawful route for output, installation or operation.
Technical	Claim depends on uncontrolled data or unstable configuration.
Commercial	No representative baseline, buyer commitment or viable economics.
Manufacturing	Single-point supplier, incomplete product definition or unknown COGS.
Service	Failure modes cannot be diagnosed or supported at target geography.
IP / data	Ownership, access, publication or supplier rights are unresolved.
Reputation	Public wording exceeds verified scope or hides significant limitation.

13. Roles and decision rights

Small teams still need separation between doing the work, verifying the result and authorizing exposure. One person may hold several roles, but conflicts should be visible and compensated by independent review where risk is high.

Role	Decision accountability
Programme owner	Priorities, resources, integrated schedule and gate request.
Technical authority	Performance basis, configuration and technical acceptance.
Quality / evidence custodian	Document control, traceability, deviations and record integrity.
Safety / conformity lead	Risk file, approval route and critical protective evidence.
Commercial owner	Customer baseline, contract, economics and conversion.
Operations / service owner	Installability, training, reliability and fleet support.
Executive gate authority	Proceed, condition, redirect or stop based on whole-system evidence.
Independent specialist	External validation where competence, law or claim significance requires it.

14. Readiness dashboard without false precision

A dashboard should expose the weakest dimension and the evidence behind each status. A single percentage score can create unjustified confidence and should not replace gate judgment.

Status	Meaning	Required action
Grey — undefined	Scope, owner or acceptance criterion is missing.	Define before execution.
Red — blocked	Critical evidence failed or a prerequisite is absent.	Stop, contain and recover.
Amber — incomplete	Work is active; uncertainty or open action remains.	Close evidence gap before gate.
Green — accepted	Criterion met for stated scope and approved configuration.	Maintain and monitor.
Blue — externally confirmed	Independent confirmation exists for stated scope.	Control claim and validity.
Black — invalidated	Change or new evidence makes prior acceptance unreliable.	Withdraw claim and reassess.

NO AVERAGING

If a critical safety, regulatory or data-rights condition is red, overall readiness is red for the affected decision—even when other dimensions are strong.

15. Evidence room architecture

An acquisition-grade evidence room should let a qualified reviewer move from claim to underlying record without reconstructing company history from email. Public papers can describe the system; confidential evidence stays access-controlled.

Folder	Controlled contents
00 Index / status	Master index, configuration map, owners, gaps and gate history.
01 Corporate / ownership	Entity, cap table, assignments, contracts and governance.
02 Product / IP	Design baseline, software, know-how register and freedom-to-operate work.
03 Validation	Protocols, raw data, laboratory files, analyses and reports.
04 Regulatory / safety	Pathway decisions, risk files, approvals and correspondence.
05 Commercial	Pipeline, pilot agreements, baselines, pricing and conversion evidence.
06 Manufacturing	BOM, suppliers, quality records, costs, capacity and FAT.
07 Deployment / service	Site files, SAT, fleet, reliability, warranty and CAPA.
08 Financial / risk	Forecast basis, unit economics, insurance, claims and material risks.

CONFIDENTIALITY BOUNDARY

Do not publish customer data, proprietary recipes, detailed cost structures, negotiating strategy, personal information or security-sensitive design material in the public repository.

16. Ninety-day implementation sequence

The framework can begin without waiting for every document to be perfect. The first cycle should establish control of one production-representative configuration and one priority use case.

Period	Priority output
Days 1–15	Approve claim register, target use case, configuration baseline and integrated gap register.
Days 16–30	Freeze validation protocols, output sampling plan, safety actions and data architecture.
Days 31–45	Close critical design/document gaps; qualify instruments, laboratory and suppliers.
Days 46–60	Execute controlled technical runs; capture raw data, deviations and samples.
Days 61–75	Complete analysis, independent results, product updates and pilot gate review.
Days 76–90	Contract the flagship pilot only if site, output, safety, data and commercial prerequisites pass.

DISCIPLINE

Schedule pressure is not evidence. If a critical prerequisite fails, record the decision and recover the gap rather than relabeling the activity as a successful pilot.

17. Scale-up and strategic-review triggers

Scale, partnership and transaction discussions should be triggered by evidence maturity, not by the mere passage of time. The trigger should also identify what remains uncertain.

Trigger	Review question
Repeated technical performance	Is the operating window reproducible on the priority feedstock?
Qualified output pathway	Can output be handled lawfully at the target sites and volumes?
Paid customer conversion	Does measured value support an acceptable commercial model?
Reproducible product build	Can equivalent units pass release without founder-dependent rework?
Safe deployment	Are conformity, site, training and commissioning controls repeatable?
Fleet reliability	Do availability and service cost support expansion?
Protected knowledge / data	Can rights and know-how transfer cleanly under diligence?
Evidence-room completeness	Can material claims be traced to current, controlled records?

Conclusion

The Rapid Organic white-paper series defines the architecture of a credible waste-to-value programme. WP-012 turns that architecture into a controlled sequence of claims, evidence, gates and decisions. Its value is not in creating another layer of paperwork, but in preventing premature scale, ambiguous claims and fragmented proof.

The immediate programme priority is one integrated readiness baseline for the production-representative configuration and priority feedstock: claim register, configuration map, evidence index, risk register, gate status and named owners. From that baseline, each test, pilot, supplier decision and site deployment should close a defined gap.

DECISION PRINCIPLE

Advance when the evidence justifies the next exposure. Stop or redirect when a critical condition is unproven—even if the technology appears promising.

Scope and limitations

This public framework is general programme guidance. It does not state that Rapid Organic has completed every activity described, guarantee performance or commercial results, provide legal or regulatory approval, certify equipment, replace site-specific engineering, or disclose confidential corporate, customer, financial or intellectual-property information.

Series references

Rapid Organic public knowledge base — WP-001 through WP-011

ISO 9001 — Quality management systems

ISO 12100 — Safety of machinery: risk assessment and risk reduction

Government of Canada — Waste management resources

Canadian Centre for Occupational Health and Safety